

13.5 Solution: (a) In the particle's rest frame, the screened Coulomb potential is

$$V(t', \mathbf{x}') = ze \frac{e^{-k_D r'}}{r'},$$

where $\mathbf{x}' = (x', y', z')$, and

$$r' = \sqrt{x'^2 + y'^2 + z'^2}.$$

Assuming that the particle is moving nonrelativistically along the x -direction in a straight line with velocity v , then the particle's coordinates expressed with coordinates of the lab frame is $\mathbf{x}' = (x - vt, y, z)$, and the screened Coulomb potential in the lab frame is given by

$$V(t, \mathbf{x}) = ze \frac{e^{-k_D r}}{r},$$

with

$$r' = \sqrt{(x - vt)^2 + y^2 + z^2},$$

by Galileo transform.

The Fourier transform of the screened potential in the lab frame is

$$\begin{aligned} \Phi(\mathbf{k}, \omega) &= \frac{1}{(2\pi)^2} \int d^3x \int dt V(t, \mathbf{x}) e^{i\omega t - i\mathbf{k} \cdot \mathbf{x}} \\ &= \frac{ze}{2\pi} \delta(\omega - \mathbf{k} \cdot \mathbf{v}) \int d^3x \frac{e^{-k_D r} e^{-i\mathbf{k} \cdot \mathbf{x}}}{r} \\ &= \frac{2ze}{k^2 + k_D^2} \delta(\omega - \mathbf{k} \cdot \mathbf{v}), \end{aligned}$$

where we have used the well-know result that

$$\int \frac{e^{-k_D r} e^{-i\mathbf{k} \cdot \mathbf{x}}}{r} d^3x = \frac{4\pi}{k^2 + k_D^2}.$$

Using Eq. (13.26) and noticing that the particle moves nonrelativistically, $v \ll c$, the electric field is given by $\mathbf{E}(\mathbf{k}, \omega) = -ik\Phi(\mathbf{k}, \omega)$, and by Eq. (13.28),

$$\mathbf{E}(\omega) = \frac{1}{(2\pi)^{3/2}} \int \mathbf{E}(\mathbf{k}, \omega) e^{ik_2 b} d^3k.$$

For the component of $\mathbf{E}$ parallel to $\mathbf{v}$, we have

$$\begin{aligned} E_1(\omega) &= \frac{-2ize}{(2\pi)^{3/2}} \int \frac{k_1 \delta(\omega - k_1 v)}{k^2 + k_D^2} e^{ik_2 b} d^3k \\ &= \frac{-2ize\omega}{(2\pi)^{3/2} v^2} \int_{-\infty}^{+\infty} dk_2 e^{ik_2 b} \int_{-\infty}^{+\infty} \frac{dk_3}{k_2^2 + k_3^2 + \lambda^2}, \end{aligned}$$

with $\lambda^2 = k_D^2 + \omega^2/v^2$. Perform the integrals in the transverse directions as in Section 13.3, we have

$$E_1(\omega) = \frac{-2ize\omega}{(2\pi)^{3/2} v^2} \int_{-\infty}^{+\infty} \frac{\pi}{(k_2^2 + \lambda^2)^{1/2}} e^{ik_2 b} dk_2 = -\frac{ize\omega}{v^2} \sqrt{\frac{2}{\pi}} K_0(\lambda b).$$

For the transverse component,

$$\begin{aligned}
E_2(\omega) &= \frac{-2ize}{(2\pi)^{3/2}} \int \frac{k_2 \delta(\omega - k_1 v)}{k^2 + k_D^2} e^{ik_2 b} d^3 k \\
&= \frac{-2ize\omega}{(2\pi)^{3/2} v} \int_{-\infty}^{+\infty} \frac{\pi k_2}{(k_2^2 + \lambda^2)^{1/2}} e^{ik_2 b} dk_2 \\
&= \frac{ze}{v} \sqrt{\frac{2}{\pi}} \lambda K_1(\lambda b).
\end{aligned}$$

Following the same procedure as in Problem 13.1 (b), the transverse momentum transferred to the electron is

$$\Delta p = \int e E_2(t) dt = \sqrt{2\pi} e E_2(\omega = 0).$$

Therefore,

$$\Delta p = \sqrt{2\pi} \cdot e \cdot \frac{ze}{v} \sqrt{\frac{2}{\pi}} k_D K_1(k_D b) = \frac{2ze^2}{v} k_D K_1(k_D b),$$

and the energy transfer is

$$\Delta E(b) = \frac{(\Delta p)^2}{2m} = \frac{2z^2 e^4}{mv^2} k_D^2 K_1^2(k_D b).$$

(b) Using Eq. (13.35), we have

$$\left(\frac{dE}{dx} \right)_{k_D b < 1} = 2\pi N Z \int_0^{k_D^{-1}} \Delta E(b) b db = \frac{z^2 e^2}{v^2} \frac{4\pi N Z e^2}{m} \int_0^1 x K_1(x)^2 dx.$$

Since $K_1(x) \sim 1/x$ for $x \rightarrow 0$, the integral diverges at the lower limit. To remedy this, we can separate the divergent part of the integral and set a cutoff. Then,

$$\begin{aligned}
\int_0^1 x K_1(x)^2 dx &= \int_0^1 x \left(\frac{1}{x} + K_1(x) - \frac{1}{x} \right)^2 dx = \int_0^1 \left[\frac{1}{x} + 2 \left(K_1(x) - \frac{1}{x} \right) + x \left(K_1(x) - \frac{1}{x} \right)^2 \right] dx \\
&\rightarrow \int_{k_D b_{\min}}^1 \frac{dx}{x} + \int_0^1 (1 + x K_1(x)) \left(K_1(x) - \frac{1}{x} \right) dx \\
&= \log \left(\frac{1}{k_D b_{\min}} \right) - 0.54497 = \log \left(\frac{1}{1.72 k_D b_{\min}} \right),
\end{aligned}$$

and

$$\left(\frac{dE}{dx} \right)_{k_D b < 1} = \frac{z^2 e^2}{v^2} \omega_p^2 \log \left(\frac{1}{1.72 k_D b_{\min}} \right),$$

which differs Jackson's result with a constant. This is due to the different regularization scheme used here, compared with Jackson.